MedNLP - Clinical Text Understanding

Project Overview

MedNLP is a clinical natural language processing system designed to extract structured information from unstructured medical records. The system performs named entity recognition (NER), relation extraction, and clinical coding to automate the processing of over 15,000 medical documents daily.

Problem Statement

Healthcare organizations generate millions of unstructured clinical notes annually. Manual processing is time-consuming, expensive, and prone to errors. MedNLP automates extraction of diagnoses, medications, procedures, and their relationships from clinical text, enabling downstream analytics and decision support systems.

Technical Approach

Model Architecture:
- Base: ClinicalBERT (BioBERT fine-tuned on MIMIC-III clinical notes)
- NER: Token classification head with CRF layer
- Relation Extraction: Entity-aware attention mechanism
- Clinical Coding: Multi-label classification for ICD-10 codes

Training Data:
- MIMIC-III/IV clinical notes (de-identified)
- i2b2/VA challenge datasets for NER
- Custom annotated dataset of 5,000 discharge summaries

Entity Types Extracted:
- Problem: Diseases, symptoms, diagnoses
- Treatment: Medications, procedures, therapies
- Test: Lab results, imaging studies
- Person: Patients, providers
- Temporal: Dates, durations, frequencies

Relation Types:
- Treats: Medication treats condition
- Causes: Condition causes symptom
- Indicates: Test indicates diagnosis
- Administers: Provider administers treatment

Implementation

Data Pipeline:
- De-identification module for PHI removal
- Text preprocessing (section detection, abbreviation expansion)
- Annotation pipeline with Prodigy for active learning

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Model Training:
- Mixed precision training with PyTorch Lightning
- Gradient accumulation for large batch sizes
- Early stopping based on validation F1

Deployment:
- ONNX Runtime for optimized inference
- Docker containers on Kubernetes
- Batch processing for non-real-time workloads
- REST API for real-time queries

Results

NER Performance:
- Problem entities: F1 = 0.94
- Treatment entities: F1 = 0.92
- Test entities: F1 = 0.91
- Overall NER F1: 0.93

Relation Extraction:
- Treats relations: F1 = 0.89
- Causes relations: F1 = 0.86
- Overall RE F1: 0.88

Clinical Coding:
- Micro F1: 0.81
- Macro F1: 0.74
- Top-10 accuracy: 0.92

Impact

- Automated processing of 15,000+ documents daily
- Reduced manual chart review time by 70%
- Enabled real-time clinical decision support alerts
- Improved quality metrics reporting accuracy by 35%